

Lecture 9

Adjoint; Self-Adjoint, Unitary, and Normal Operators

The Riesz theorem of Lecture 8 turned vectors into functionals. Applied to operators it produces the *adjoint* $T^\dagger$, the abstract version of the conjugate transpose, defined by $\langle T^\dagger u, v \rangle = \langle u, Tv \rangle$. With the adjoint in hand we can single out the three families of operators that organize the finite-dimensional operator model used in quantum mechanics: the *self-adjoint* operators $T^\dagger = T$, whose real eigenvalues label the possible outcomes of an ideal projective measurement; the *unitary* operators $U^\dagger = U^{-1}$, which preserve inner products and implement closed-system time evolution and unitary symmetries (other symmetries may be antiunitary); and the *normal* operators $T^\dagger T = TT^\dagger$, the widest class that admits an orthonormal eigenbasis in finite-dimensional complex spaces. This lecture builds the adjoint and proves the spectral facts—real eigenvalues, orthogonal eigenvectors, unit-circle spectra—that the spectral theorem of Lecture 10 will complete.

Learning objectives

By the end of this lecture you will be able to:

- ▶ Compute the adjoint $T^\dagger$ and use $\langle T^\dagger u, v \rangle = \langle u, Tv \rangle$ and its algebra.
- ▶ Classify operators as self-adjoint, unitary, or normal.
- ▶ Prove the core facts: real eigenvalues, unit-circle spectrum, orthogonal eigenvectors.
- ▶ In the finite/discrete pure-point model, identify observables, unitary versus antiunitary symmetries, and the class the finite spectral theorem diagonalizes.

1 The adjoint

Definition 9.1 (Adjoint). Let $T \in \mathcal{L}(V)$ on a finite-dimensional inner product space. The *adjoint* $T^\dagger$ is the unique operator satisfying

$$\langle T^\dagger u, v \rangle = \langle u, Tv \rangle \quad \text{for all } u, v \in V.$$

Existence and uniqueness are Riesz in disguise: for fixed u , the map $v \mapsto \langle u, Tv \rangle$ is a linear functional, so equals $\langle w, v \rangle$ for a unique w_u . Uniqueness also proves that these representatives depend linearly on u : $w_{u_1+u_2} = w_{u_1} + w_{u_2}$ because both sides represent $v \mapsto \langle u_1 + u_2, Tv \rangle$. For $a \in \mathbb{F}$, the functional for au is $\phi_{au} = \overline{a} \phi_u$, while $\langle aw_u, v \rangle = \overline{a} \langle w_u, v \rangle$; uniqueness therefore gives $w_{au} = aw_u$. Thus setting $T^\dagger u = w_u$ really does define a linear operator. Equivalently, taking conjugates, $\langle Tu, v \rangle = \langle u, T^\dagger v \rangle$: the adjoint moves an operator across the inner product from one slot to the other.

Proposition 9.2 (The adjoint is the conjugate transpose). In an orthonormal basis, $\mathcal{M}(T^\dagger) = \mathcal{M}(T)^\dagger$: the matrix of the adjoint is the conjugate transpose of the matrix of T .

Proof. With orthonormal $e_1, \dots, e_n$, the (j, k) entry of T is $T_{jk} = \langle e_j, Te_k \rangle$. Hence

$$(T^\dagger)_{jk} = \langle e_j, T^\dagger e_k \rangle = \langle Te_j, e_k \rangle = \overline{\langle e_k, Te_j \rangle} = \overline{T_{kj}},$$

which is exactly the conjugate transpose. ■

Proposition 9.3 (Algebra of adjoints). For $S, T \in \mathcal{L}(V)$ and $a \in \mathbb{F}$: $(S + T)^\dagger = S^\dagger + T^\dagger$, $(aT)^\dagger = \bar{a} T^\dagger$, $(ST)^\dagger = T^\dagger S^\dagger$, $(T^\dagger)^\dagger = T$, and $I^\dagger = I$. If T is invertible then $(T^{-1})^\dagger = (T^\dagger)^{-1}$.

Proof. Each follows by checking the defining relation. For the product, $\langle (ST)^\dagger u, v \rangle = \langle u, STv \rangle = \langle S^\dagger u, Tv \rangle = \langle T^\dagger S^\dagger u, v \rangle$ for all v , so $(ST)^\dagger = T^\dagger S^\dagger$. The conjugate on a comes from conjugate-linearity in the first slot. ■

Theorem 9.4 (Null space and range). For $T \in \mathcal{L}(V)$, $\text{null } T^\dagger = (\text{range } T)^\perp$ and $\text{range } T^\dagger = (\text{null } T)^\perp$.

Proof. $u \in \text{null } T^\dagger$ iff $T^\dagger u = 0$ iff $\langle T^\dagger u, v \rangle = 0$ for all v iff $\langle u, Tv \rangle = 0$ for all v iff $u \perp \text{range } T$. That is the first identity. Apply it now to $T^\dagger$: $\text{null } T = (\text{range } T^\dagger)^\perp$, since $(T^\dagger)^\dagger = T$. Taking orthogonal complements and using the finite-dimensional identity $W^{\perp\perp} = W$ gives $(\text{null } T)^\perp = \text{range } T^\dagger$, the second identity. ■

Example 9.5 (An adjoint by conjugate transpose). $A = \begin{pmatrix} 1 & 2-i \\ 3i & 4 \end{pmatrix}$ has adjoint $A^\dagger = \begin{pmatrix} 1 & -3i \\ 2+i & 4 \end{pmatrix}$: transpose, then conjugate. Since $A^\dagger \neq A$ and $A^\dagger \neq A^{-1}$, this operator is neither self-adjoint nor unitary. ◀

Example 9.6 (Adjoint of the shift). The right shift $T(z_1, \dots, z_n) = (0, z_1, \dots, z_{n-1})$ on $\mathbb{C}^n$ has adjoint the left shift $T^\dagger(z_1, \dots, z_n) = (z_2, \dots, z_n, 0)$: with the boundary convention $u_0 = 0$, from $\langle Tu, v \rangle = \sum_k \overline{u_{k-1}} v_k = \sum_j \overline{u_j} v_{j+1} = \langle u, T^\dagger v \rangle$ the adjoint reverses the shift direction. ◀

Example 9.7 (Adjoint of a product reverses order). For $S = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $T = \begin{pmatrix} 1 & 0 \\ i & 0 \end{pmatrix}$, $ST = \begin{pmatrix} i & 0 \\ 0 & 0 \end{pmatrix}$, so $(ST)^\dagger = \begin{pmatrix} -i & 0 \\ 0 & 0 \end{pmatrix} = T^\dagger S^\dagger$. Adjunction reverses the order of a product, just as inversion does. ◀

Example 9.8 (The defining relation, checked). With $T = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $u = (1, 0)$, $v = (0, 1)$: $\langle u, Tv \rangle = \langle (1, 0), (1, 0) \rangle = 1$, while $T^\dagger u = (0, 1)$ gives $\langle T^\dagger u, v \rangle = \langle (0, 1), (0, 1) \rangle = 1$. The two sides of $\langle T^\dagger u, v \rangle = \langle u, Tv \rangle$ agree, as the definition requires. ◀

Remark 9.9 (Detecting the zero operator). $T = 0 \iff T^\dagger = 0 \iff T^\dagger T = 0$. The last equivalence is handy because $T^\dagger T$ is self-adjoint with $\langle v, T^\dagger T v \rangle = \|Tv\|^2$, so $T^\dagger T = 0$ forces $Tv = 0$ for every v .

Corollary 9.10 (Injectivity and surjectivity). T is injective iff $T^\dagger$ is surjective, and T is surjective iff $T^\dagger$ is injective; hence T is invertible iff $T^\dagger$ is.

Proof. T injective $\iff \text{null } T = \{0\} \iff (\text{range } T^\dagger)^\perp = \{0\} \iff \text{range } T^\dagger = V$ by Theorem 9.4, i.e. $T^\dagger$ surjective. Exchanging T and $T^\dagger$ gives the other statement. ■

2 Self-adjoint operators

Definition 9.11 (Self-adjoint). $T \in \mathcal{L}(V)$ is *self-adjoint* (*Hermitian*) if $T^\dagger = T$, i.e. $\langle Tu, v \rangle = \langle u, Tv \rangle$ for all u, v . Its matrix in an orthonormal basis is a Hermitian matrix, $A^\dagger = A$.

In the finite/discrete pure-point model these operators encode observables. Two facts make them useful there.

Theorem 9.12 (Real eigenvalues). Every eigenvalue of a self-adjoint operator is real.

Proof. Let $Tv = \lambda v$ with $v \neq 0$. Then $\lambda \|v\|^2 = \langle v, Tv \rangle = \langle Tv, v \rangle = \bar{\lambda} \|v\|^2$, using self-adjointness in the middle step. Since $\|v\|^2 \neq 0$, $\lambda = \bar{\lambda}$ is real. ■

Both of the next two theorems turn on one small fact, which is worth isolating rather than smuggling into a proof.

Lemma 9.13 (A vanishing quadratic form). Let $S \in \mathcal{L}(V)$ satisfy $\langle v, Sv \rangle = 0$ for every $v \in V$. If $\mathbb{F} = \mathbb{C}$, then $S = 0$. If $\mathbb{F} = \mathbb{R}$, the same follows provided S is self-adjoint.

Proof. Expanding $0 = \langle u + v, S(u + v) \rangle$ and using $\langle u, S \rangle u = \langle v, S \rangle v = 0$ gives

$$\langle u, Sv \rangle = -\langle v, Su \rangle \quad \text{for all } u, v. \quad (9.1)$$

Over $\mathbb{C}$, replace u by iu in (9.1). The left side becomes $\bar{i} \langle u, S \rangle v = -i \langle u, S \rangle v$ (conjugate-linear first slot) and the right side $-i \langle v, S \rangle u$ (linear second slot), so $\langle u, S \rangle v = \langle v, S \rangle u$. Together with (9.1) this forces $\langle u, S \rangle v = 0$ for all u, v , hence $Sv = 0$ for every v . Over $\mathbb{R}$ with S self-adjoint, $\langle v, S \rangle u = \langle Sv, u \rangle = \langle u, Sv \rangle$, and (9.1) reads $\langle u, S \rangle v = -\langle u, S \rangle v$, giving the same conclusion. ■

Theorem 9.14 (Self-adjoint via a real quadratic form). For $\mathbb{F} = \mathbb{C}$, an operator T is self-adjoint if and only if $\langle v, Tv \rangle \in \mathbb{R}$ for every v .

Proof. If $T^\dagger = T$ then $\langle v, Tv \rangle = \langle Tv, v \rangle = \overline{\langle v, Tv \rangle}$ is real. Conversely, if $\langle v, Tv \rangle$ is always real then $\langle v, Tv \rangle = \overline{\langle v, Tv \rangle} = \langle Tv, v \rangle = \langle v, T^\dagger v \rangle$, so $\langle v, (T - T^\dagger)v \rangle = 0$ for all v , so $T = T^\dagger$ by Lemma 9.13. ■

Theorem 9.15 (Orthogonal eigenvectors). Eigenvectors of a self-adjoint operator for distinct eigenvalues are orthogonal.

Proof. Let $Tu = \alpha u$, $Tv = \beta v$ with $\alpha \neq \beta$ (both real). Then

$$\alpha \langle u, v \rangle = \langle Tu, v \rangle = \langle u, Tv \rangle = \beta \langle u, v \rangle,$$

using $\langle Tu, v \rangle = \bar{\alpha} \langle u, v \rangle = \alpha \langle u, v \rangle$ on the left. Thus $(\alpha - \beta) \langle u, v \rangle = 0$, forcing $\langle u, v \rangle = 0$. ■

In the finite ideal-projective model, a measurement of a self-adjoint observable returns one of its real eigenvalues. Distinct eigenvalues have orthogonal eigenspaces and can therefore be distinguished by the corresponding spectral measurement. Continuous-spectrum observables require the projection-valued measures of Lecture 14 instead of ordinary normalizable eigenvectors.

Example 9.16 (The Pauli matrices are observables). Each Pauli matrix $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ equals its own conjugate transpose, so all three are self-adjoint with real eigenvalues ± 1 —the two outcomes of a spin measurement—and orthogonal eigenstates. ◀

Example 9.17 (Orthogonal eigenvectors in action). The Hermitian matrix $A = \begin{pmatrix} 2 & i \\ -i & 2 \end{pmatrix}$ has real eigenvalues 1, 3 with eigenvectors $(1, i)$ and $(1, -i)$. Their inner product $\langle (1, i), (1, -i) \rangle = 1 \cdot 1 + i \cdot (-i) = 1 + (-i)(-i) = 1 - 1 = 0$ confirms Theorem 9.15. ◀

Example 9.18 (An orthonormal eigenbasis). Normalizing those eigenvectors gives the orthonormal pair $e_1 = \frac{1}{\sqrt{2}}(1, i)$, $e_2 = \frac{1}{\sqrt{2}}(1, -i)$, in which the operator becomes $\text{diag}(1, 3)$. A self-adjoint operator is thus diagonal in an orthonormal basis—the spectral theorem of Lecture 10, previewed. ◀

Example 9.19 (A 3×3 self-adjoint operator). The real symmetric matrix $\begin{pmatrix} 14 & -13 & 8 \\ -13 & 14 & 8 \\ 8 & 8 & -7 \end{pmatrix}$ equals its transpose, hence is self-adjoint, with real eigenvalues 27, 9, -15 and orthonormal eigenvectors $\frac{1}{\sqrt{2}}(1, -1, 0)$, $\frac{1}{\sqrt{3}}(1, 1, 1)$, $\frac{1}{\sqrt{6}}(1, 1, -2)$. In that orthonormal basis it is $\text{diag}(27, 9, -15)$ —real spectrum, orthogonal eigenstates, exactly as the theory predicts. ◀

Remark 9.20 (The real case). A real symmetric matrix is self-adjoint for the real dot product, so it too has real eigenvalues and an *orthogonal* eigenbasis—the real spectral theorem. This is why the principal axes of an ellipsoid, the moments of inertia of a rigid body, and the normal modes of a coupled system come out mutually perpendicular.

Example 9.21 (Diagonalizing σ_x). $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has real eigenvalues ± 1 with orthonormal eigenvectors $|\pm\rangle = \frac{1}{\sqrt{2}}(1, \pm 1)$. In the unitary basis $U = (|+\rangle |-\rangle)$,

$$U^\dagger \sigma_x U = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma_z,$$

so the Hadamard change of basis turns σ_x into σ_z : a self-adjoint operator becomes a real diagonal matrix in its orthonormal eigenbasis. ◀

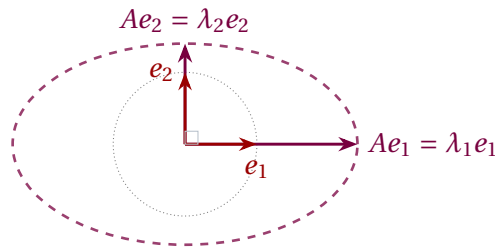


Figure 1: A self-adjoint operator stretches along an *orthogonal* eigenbasis $e_1 \perp e_2$ by real factors λ_1, λ_2 ; the unit circle maps to a possibly degenerate ellipse (a segment or point when an eigenvalue vanishes) with axes along the eigendirections.

Remark 9.22 (Products of observables). The product of two self-adjoint operators is self-adjoint if and only if they commute, since $(ST)^\dagger = T^\dagger S^\dagger = TS$. A product of observables is again an observable only for compatible (commuting) observables.

Example 9.23 (Expectation values are real). For a self-adjoint A and a normalized state $|\psi\rangle$ in the finite model, the expectation value $\langle A \rangle = \langle \psi | A | \psi \rangle = \langle \psi, A\psi \rangle$ is real by [Theorem 9.14](#). The measured average of an observable is therefore a real number, exactly as physics requires. ◀

Example 9.24 (Energies are real, eigenstates orthogonal). For a finite-dimensional, time-independent Hamiltonian $\hat{H}$, the energy eigenvalues E_n are real and eigenspaces for distinct energies are orthogonal. Choose a normalized orthonormal basis inside every degenerate eigenspace; only then does $\langle E_m | E_n \rangle = \delta_{mn}$ hold for the chosen vectors. Equivalently, write a state as $|\psi\rangle = \sum_E P_E |\psi\rangle$ and evolve each whole energy eigenspace by $e^{-iEt/\hbar}$. Discrete pure-point models work the same way; continuous-spectrum Hamiltonians such as the free Hamiltonian require Lecture 14. ◀

Example 9.25 (A real quadratic form). For $\sigma_z = \text{diag}(1, -1)$ and $v = (a, b)$, $\langle v, \sigma_z v \rangle = \bar{a}a - \bar{b}b = |a|^2 - |b|^2 \in \mathbb{R}$, illustrating **Theorem 9.14**: the quadratic form of an observable is always real, whatever the state. $\triangleleft$

3 Unitary operators

Definition 9.26 (Unitary). $U \in \mathcal{L}(V)$ is *unitary* if $U^\dagger U = UU^\dagger = I$, i.e. $U^\dagger = U^{-1}$.

Theorem 9.27 (Unitary means inner-product preserving). For $U \in \mathcal{L}(V)$ the following are equivalent:

1. U is unitary;
2. $\langle Uu, Uv \rangle = \langle u, v \rangle$ for all u, v (preserves the inner product);
3. $\|Uv\| = \|v\|$ for all v (preserves norms);
4. U maps some (every) orthonormal basis to an orthonormal basis.

Proof. (1) $\Rightarrow$ (2): $\langle Uu, Uv \rangle = \langle u, U^\dagger Uv \rangle = \langle u, v \rangle$. (2) $\Rightarrow$ (3): take $u = v$. (3) $\Rightarrow$ (2): recover the inner product from the norm by polarization (Lecture 7). (2) $\Rightarrow$ (4): if $\langle e_j, e_k \rangle = \delta_{jk}$ then $\langle Ue_j, Ue_k \rangle = \delta_{jk}$. (4) $\Rightarrow$ (1): if U sends the ONB e_k to an ONB, then $U^\dagger U$ has matrix δ_{jk} , so $U^\dagger U = I$, and in finite dimensions a left inverse is a two-sided inverse. $\blacksquare$

Theorem 9.28 (Unit-circle spectrum). Every eigenvalue of a unitary operator has absolute value 1.

Proof. If $Uv = \lambda v$ with $v \neq 0$, then $\|v\| = \|Uv\| = |\lambda| \|v\|$, so $|\lambda| = 1$; the eigenvalues lie on the unit circle, $\lambda = e^{i\theta}$. $\blacksquare$

On a complex finite-dimensional space, time evolution $e^{-i\hat{H}t/\hbar}$ is unitary, which is why it preserves total probability $\|\psi\| = 1$; unitary symmetry operations preserve the same inner products.

Example 9.29 (Unitary operators). The rotation $R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ satisfies $R^\dagger R = I$ and preserves lengths in $\mathbb{R}^2$. Each Pauli matrix is also unitary (besides being Hermitian), since $\sigma^\dagger \sigma = \sigma^2 = I$; indeed a Hermitian operator is unitary precisely when $A^2 = I$. (Self-inverse alone is not enough: $\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$ squares to I without being Hermitian.) $\triangleleft$

Example 9.30 (Unit-circle eigenvalues). Over $\mathbb{C}$ the rotation R above has eigenvalues $e^{\pm i\theta}$ with eigenvectors $(1, \mp i)$: both lie on the unit circle, as **Theorem 9.28** requires, and the eigenvectors are orthogonal because R is normal. $\triangleleft$

Example 9.31 (Verifying unitarity). The Hadamard operator $U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ satisfies $U^\dagger U = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = I$, so it is unitary; it maps the basis $|0\rangle, |1\rangle$ to the orthonormal pair $|\pm\rangle$ of Lecture 7. $\triangleleft$

Example 9.32 (Unitarity preserves overlaps). Under the same Hadamard U , $|0\rangle \mapsto |+\rangle$ and $|1\rangle \mapsto |-\rangle$, and $\langle U0, U1 \rangle = \langle + | - \rangle = 0 = \langle 0 | 1 \rangle$: the overlap of the images equals the overlap of the originals, as **Theorem 9.27** guarantees. $\triangleleft$

Remark 9.33 (Change of orthonormal basis is unitary). The matrix relating two orthonormal bases has orthonormal columns, hence is unitary (**Theorem 9.27**). Passing between orthonormal bases is always a unitary transformation, and an operator's matrix transforms

as $A' = U^\dagger A U$ with U unitary—the inner-product-respecting refinement of the similarity $A' = P^{-1} A P$ from Lecture 4.

Remark 9.34 (Extension: generating unitaries). On a complex inner-product space, if A is self-adjoint then e^{iA} is unitary, because $(e^{iA})^\dagger = e^{-iA^\dagger} = e^{-iA} = (e^{iA})^{-1}$. This is the finite-dimensional shadow of Stone's theorem: every *strongly continuous* one-parameter unitary group has a unique self-adjoint generator. Ordinary continuity suffices in finite dimensions; unbounded generators and their domains belong to Lecture 14.

4 Normal operators

Definition 9.35 (Normal). $T \in \mathcal{L}(V)$ is *normal* if it commutes with its adjoint, $T^\dagger T = T T^\dagger$.

Self-adjoint operators ($T^\dagger = T$) and unitary operators ($T^\dagger = T^{-1}$) are obviously normal; normality is the common umbrella. Its usefulness rests on one identity.

Theorem 9.36 (Normal length identity). T is normal if and only if $\|T v\| = \|T^\dagger v\|$ for all v .

Proof. For any T , $\|T v\|^2 = \langle T v, T v \rangle = \langle v, T^\dagger T v \rangle$ and $\|T^\dagger v\|^2 = \langle v, T T^\dagger v \rangle$. These agree for all v iff $\langle v, (T^\dagger T - T T^\dagger) v \rangle = 0$ for all v ; as $T^\dagger T - T T^\dagger$ is self-adjoint, **Lemma 9.13** says that holds iff it is 0. ■

Corollary 9.37 (Shared eigenvectors with the adjoint). If T is normal, then $T v = \lambda v$ if and only if $T^\dagger v = \bar{\lambda} v$. Consequently eigenvectors of T for distinct eigenvalues are orthogonal.

Proof. $T - \lambda I$ is normal (it commutes with its adjoint $T^\dagger - \bar{\lambda} I$), so by **Theorem 9.36** $\|(T - \lambda I) v\| = \|(T^\dagger - \bar{\lambda} I) v\|$; the left side is 0 iff the right side is. For orthogonality, with $T u = \alpha u$ and $T v = \beta v$ we get $T^\dagger v = \bar{\beta} v$, so $\bar{\alpha} \langle u, v \rangle = \langle T u, v \rangle = \langle u, T^\dagger v \rangle = \bar{\beta} \langle u, v \rangle$, forcing $\langle u, v \rangle = 0$ when $\alpha \neq \beta$. ■

Example 9.38 (Normal but neither Hermitian nor unitary). $A = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ has $A^\dagger = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ and $A A^\dagger = A^\dagger A = 2I$, so A is normal; yet $A^\dagger \neq A$ and $A A^\dagger = 2I \neq I$, so it is neither self-adjoint nor unitary. Its eigenvalues $1 \pm i$ are neither real nor on the unit circle—the general normal case. ◀

Example 9.39 (Checking normality by lengths). For the same A , $\|A v\|^2 = \langle v, A^\dagger A v \rangle = 2\|v\|^2 = \langle v, A A^\dagger v \rangle = \|A^\dagger v\|^2$, so $\|A v\| = \|A^\dagger v\|$ for every v —the length test of **Theorem 9.36**, passed. ◀

Remark 9.40 (Positive operators). For any T , the operator $T^\dagger T$ is self-adjoint and *positive*: $\langle v, T^\dagger T v \rangle = \|T v\|^2 \geq 0$, so its eigenvalues are nonnegative. Positive operators are the analogue of the nonnegative reals and underlie the singular value decomposition and quantum density matrices.

Example 9.41 (Orthogonal eigenvectors of a normal operator). The normal operator $A = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ has eigenvalues $1 \pm i$ with eigenvectors $(1, \pm i)$, and $\langle (1, i), (1, -i) \rangle = 1 + (-i)(-i) = 0$. Distinct eigenvalues give orthogonal eigenvectors, as **Corollary 9.37** promises—even though A is neither self-adjoint nor unitary. ◀

Remark 9.42 (Extension: polar and singular value decompositions). Just as every complex number is $r e^{i\theta}$, every operator factors as $T = U P$ with U unitary and $P = \sqrt{T^\dagger T}$ positive

(the *polar decomposition*); diagonalizing P in orthonormal bases yields the *singular value decomposition* $T = U\Sigma V^\dagger$, the workhorse of data analysis and low-rank approximation. Both rest on the adjoint built here.

Theorem 9.43 (Cartesian decomposition). Every operator on a complex space is $T = A + iB$ with $A = \frac{1}{2}(T + T^\dagger)$ and $B = \frac{1}{2i}(T - T^\dagger)$ self-adjoint, and T is normal if and only if A and B commute.

Proof. Both A and B are self-adjoint by direct check, and $A + iB = T$. A short computation gives $AB - BA = \frac{1}{2i}(T^\dagger T - TT^\dagger)$, which vanishes exactly when T is normal. The pair (A, B) are the operator “real and imaginary parts” of T , in analogy with $z = a + ib$. ■

Example 9.44 (Cartesian parts of an operator). For $T = \begin{pmatrix} 1 & 2i \\ 0 & 3 \end{pmatrix}$, $T^\dagger = \begin{pmatrix} 1 & 0 \\ -2i & 3 \end{pmatrix}$, so the self-adjoint parts are

$$A = \frac{T + T^\dagger}{2} = \begin{pmatrix} 1 & i \\ -i & 3 \end{pmatrix}, \quad B = \frac{T - T^\dagger}{2i} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

both Hermitian, with $T = A + iB$. Here $AB \neq BA$, so T is not normal—as expected for a non-diagonal upper-triangular matrix. ◀

Remark 9.45 (Extension: skew-adjoint operators). On a complex inner-product space, an operator with $B^\dagger = -B$ is *skew-adjoint*; then iB is self-adjoint, so skew-adjoint operators have purely imaginary eigenvalues. Their exponentials are unitary, which is why the generator of time evolution, $-i\hat{H}/\hbar$, is skew-adjoint—the bridge between observables and symmetries.

Example 9.46 (A skew-adjoint operator). $B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, regarded on $\mathbb{C}^2$ (or obtained by complexifying its action on $\mathbb{R}^2$), satisfies $B^\dagger = -B$, so it is skew-adjoint with purely imaginary eigenvalues $\pm i$; indeed $iB = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \sigma_y$ is Hermitian. Its exponential $e^{\theta B}$ is precisely the rotation of **Example 9.29**: a unitary generated by a skew-adjoint operator. ◀

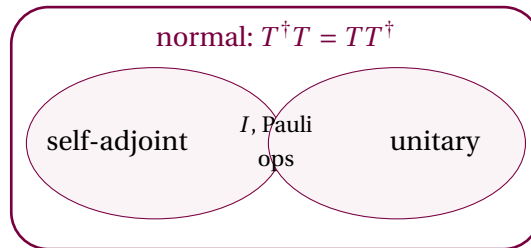


Figure 2: The self-adjoint and unitary operators are distinct families inside the normal operators; their overlap (operators that are both, such as I and the Pauli matrices) consists of self-inverse Hermitian operators.

5 Observables, symmetries, and what comes next

For the finite/discrete pure-point model, the dictionary is now complete. A self-adjoint operator plays the role of a real number: its eigenvalues are real (ideal-projective measurement outcomes) and its eigenvectors for distinct outcomes are orthogonal (distinguishable states). A unitary operator plays the role of a complex number of modulus one: it preserves all inner products and its eigenvalues are phases $e^{i\theta}$. Both are normal, and normality is exactly the property—by the finite complex spectral theorem of Lecture 10—that guarantees an orthonormal basis of eigenvectors. Multiplication by x on $L^2[0, 1]$ is the standard warning:

it is self-adjoint and normal but has no nonzero ordinary eigenvectors; Lecture 14 replaces finite sums by spectral measures.

In the finite model, self-adjoint observables have orthonormal eigenbases with real eigenvalues, and closed-system evolution is unitary. Unitary symmetries preserve inner products; antiunitary symmetries are the conjugate-linear alternative. The next lecture proves that every normal operator on a finite-dimensional complex space is diagonal in an orthonormal basis.

Proposition 9.47 (Extension: uncertainty relation). For self-adjoint A, B and a unit state $|\psi\rangle$, writing $\Delta X = X - \langle X \rangle I$ for $X = A, B$,

$$\langle (\Delta A)^2 \rangle \langle (\Delta B)^2 \rangle \geq \frac{1}{4} |\langle [A, B] \rangle|^2.$$

Proof. $\Delta A, \Delta B$ are self-adjoint, so $\langle (\Delta A)^2 \rangle = \|\Delta A|\psi\rangle\|^2$. Cauchy–Schwarz gives

$$\|\Delta A|\psi\rangle\|^2 \|\Delta B|\psi\rangle\|^2 \geq |\langle \Delta A|\psi, \Delta B|\psi \rangle|^2.$$

The imaginary part of $\langle \Delta A|\psi, \Delta B|\psi \rangle$ equals

$$\frac{1}{2i} (\langle \Delta A|\psi, \Delta B|\psi \rangle - \langle \Delta B|\psi, \Delta A|\psi \rangle) = \frac{1}{2i} \langle \psi | [A, B] | \psi \rangle,$$

using $[\Delta A, \Delta B] = [A, B]$. Since $|z|^2 \geq (\operatorname{Im} z)^2$, the bound follows. ■

Example 9.48 (Incompatible spin components). The Pauli matrices satisfy $[\sigma_x, \sigma_y] = 2i\sigma_z$, so the bound reads $\langle (\Delta\sigma_x)^2 \rangle \langle (\Delta\sigma_y)^2 \rangle \geq |\langle \sigma_z \rangle|^2$: two spin components cannot both be sharp unless the third has zero expectation. This is Heisenberg uncertainty in its barest form. ◀

Remark 9.49 (Compatible observables). Commuting finite-dimensional self-adjoint operators can be simultaneously diagonalized in a single orthonormal basis (Lecture 10). In this ideal model, *compatible* observables share a complete set of eigenstates and can hold definite values together; incompatible ones such as σ_x and σ_z cannot, which is the origin of the uncertainty relation.

Summary of main concepts

The adjoint $T^\dagger$, defined by $\langle T^\dagger u, v \rangle = \langle u, Tv \rangle$, is the conjugate transpose in an orthonormal basis and obeys $(ST)^\dagger = T^\dagger S^\dagger$, $(T^\dagger)^\dagger = T$, with null $T^\dagger = (\operatorname{range} T)^\perp$. Self-adjoint operators ($T^\dagger = T$) have real eigenvalues and orthogonal eigenvectors—the observables. Unitary operators ($U^\dagger = U^{-1}$) preserve inner products and norms, map orthonormal bases to orthonormal bases, and have eigenvalues on the unit circle—the unitary symmetries and closed-system evolutions. Antiunitary symmetries are a separate conjugate-linear possibility. Normal operators ($T^\dagger T = TT^\dagger$) satisfy $\|Tv\| = \|T^\dagger v\|$, share eigenvectors with $T^\dagger$ (conjugate eigenvalues), and have orthogonal eigenvectors for distinct eigenvalues; in finite-dimensional complex spaces they are exactly the operators the spectral theorem diagonalizes in an orthonormal basis.

- ▶ **Adjoint**— $\langle T^\dagger u, v \rangle = \langle u, Tv \rangle$; the conjugate transpose in an orthonormal basis; null $T^\dagger = (\operatorname{range} T)^\perp$.
- ▶ **Self-adjoint** ($T^\dagger = T$)—finite/discrete observable model; real eigenvalues and orthogonal eigenspaces.
- ▶ **Unitary** ($U^\dagger = U^{-1}$)—closed-system evolution and unitary symmetries; antiunitary symmetries are conjugate-linear; unitary spectrum lies on the unit circle.
- ▶ **Normal** ($T^\dagger T = TT^\dagger$)—commutes with its adjoint, so $\|Tv\| = \|T^\dagger v\|$; orthogonal eigenvectors for distinct eigenvalues, the finite complex spectral-theorem class.

Exercises for this lecture are in the companion sheet exercises-9.pdf.